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IMPACT OF TRANSPORTATION MANAGEMENT ON LOGISTICS PERFORMANCE

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ABSTRACT:

“Transportation management is considered a major element in logistics management that leads to efficient deliveries, lower costs, and customer satisfaction. In the current business world where competition prevails, businesses have started developing their interests in areas related to transportation such as route planning, vehicle use, dispatching, timing, tracking, and controlling.” This study investigates the influence of transportation management practices on logistics performance, specifically measuring delivery punctuality, operation efficiency, service quality and price effectiveness.

This research used a quantitative design using primary data from logistics workers and supply chain experts who answered questionnaires that had been structured. Visualization model, Reliability analysis, Correlation analysis and Regression analysis were performed to test the relationship of transportation management with logistics performance. There is a very high positive correlation of the behavior between efficient practices in transportation with logistics outcomes based on the findings. They maximize the reliability and cost-effectiveness of delivery while improving route optimization in real time and making scheduling more effective. This study reveals valuable knowledge for logistics managers who are seeking a means of gaining improved competitiveness through superior logistical strategy breadth.

Keywords: Transportation Management, Logistics Performance (LP), Route Optimization, Operational Efficiency (OE), Cost Reduction

INTRODUCTION

In today's highly competitive and ever-changing environment, efficient logistics has become the most critical factor for the success of any organization and for keeping up with customer expectations. The performance of logistics is mainly determined by the success of transportation management, which becomes particularly relevant since smooth transportation from point of origin to point of destination depends on its efficiency to a large degree. Economies of transportation are regarded as an extremely important element of logistics management and influence delivery reliability and efficiency in the current Indian transport market in addition to aggregate costs and service levels of the firm.

In the modern globalized world, firms have come to realize that transportation has become the critical element that cannot be neglected. Each stage of transportation management consists of a number of factors such as vehicle scheduling and routing, effective management of vehicles, cost control and application of effective IT. In practice this kind multi-task maneuvering enables companies not only to minimize their transportation expenses as well as shorten delivery times and improve resource use; it also raises the level of customer service they can offer.

This will result in delays, higher operational costs, poor coordination and reduced customer satisfaction, all of which will ultimately impact the logistics process as a whole. As a consequence, organizations have to take effective action now by adopting strategic planning techniques for transport and distribution that will both improve efficiency and retain service levels. In recent years, technological developments such as GPS tracking systems robustly increased the ability of organizations to effectively manage transportation activities at all levels (real time, live-mapping tools and real-time monitoring).

The objective of this study is to consider the influence of transportation management practices in logistics performance. It centers on the analysis of road optimization, distribution route plan, vehicle deployment and transportation cost control etc., and in-depth investigation of how those processes affect logistics performance measures which are delivery time punctuality, operation efficiency and service quality. This study will also seek to come up with an insight that will be relevant to logistics managers and decision makers in terms of improving the efficiency of their transportations, thus bettering their logistics performances overall.

LITERATURE REVIEW WITH IDENTIFIED RESEARCH GAP

The relation between logistics performance and physical distribution has already received much attention as part of the wider field of supply chain management. “As per the literature in existence, efficient transportation systems are crucial in realizing success in cost-efficiency and customer satisfaction.” The relation is different depending on the industry and the region of operation. In developed countries versus developing countries, the nature of the relation is quite unique.

As explained by the resource-based view (RBV) of the firm, the transport system of an organization can become a valuable source of competitive advantage if it is valuable, rare and non-imitable. In this connection, logistics activities associated with management practice in transport include route optimization, fleet utilization, and technology integration are thought of as strategic resources that can be used for logistics performance improvement. “Then transaction cost economics (TCE) argues that companies should try to minimize operational and coordination costs. This means that effective transportation management reduces the likelihood of uncertainties in supply chain transactions and eliminates failings.”

There are empirical studies that widely supported the positive linkage of transportation efficiency and logistics performance. “According to Christopher (2016) transportation is one of the key dimensions that influence supply chain responsiveness and cost efficiency. According to Bowersox, Closs, and Cooper (2013), well-managed transportation systems improve the reliability of delivery as well as managing inventories.” Likewise, Mentzer et al. (2001), logistics performance is therefore strongly associated with transportation capabilities, especially service quality and customer satisfaction.

Subsequent research has considered certain transportation management practices. Studies of Rodrigue, Comtois and Slack (2017) show that route optimization/network design is an important factor to shorten transit time and transportation costs. Like this in McKinnon (2018), he discovers the efficient fleet management and vehicle utilization help to improve operational efficiency and environmental sustainability. The increasing role of technology has also been broadly noted. Research by Gunasekaran et al. (2017), by using digital tools such as, GPS tracking, Transportation Management System (TMS), and real time analytics aids to increase visibility and co-ordination throughout logistics processes.

“On the other hand, there have been some studies showing mixed findings as well. While the use of information technology increases performance, the same is hindered by the increased costs and lack of technical skills,

which is often seen in most developing nations.” Likewise, other studies have shown that there can be an issue of decreased service quality and efficiency due to attempts at cost reduction in freight transport. In such a situation, the performance of transportation management will not depend on the capabilities of firms but also environmental context.

As shown through literature, poor infrastructure, strict regulations and fragmented logistics chains in India as well as other developing nations are major structural barriers. “As pointed out by different logistic performance indexes by world bank, the above barriers affect transportation management adversely. New technological developments, along with policy initiatives, have brought many new opportunities for logistics industries.” However, there have been no studies focused on transportation management in Indian firms.

Research Gap

“Despite having plenty of literature for logistics and SCM in general, there appears to be no research done concerning the relationship between logistics management in terms of transportation practices and performance of logistics in India.” Most studies vary from the fact that they have been done by looking into the logistics of only developed nations or analyzing logistics performance in a broader manner that makes it impossible to make a conclusion regarding the impact of the attributes involved in transportation.

In addition, very little literature is available as far as the evaluation of the influence of transportation management factors such as route optimization and technology usage on the logistics performance measures (e.g., delivery timely, efficient operation and service quality). This study aims to bridge this gap through conducting an empirical analysis on the impact of transportation management practices on logistics performance within the Indian firms.

This will contribute to the literature on this area and decision making process within the logistics and SCM field by conducting analysis on specific operational factors.

RESEARCH DESIGN AND METHODOLOGY

This section will provide an overview of the research methodology adopted for the analysis of the role of transportation management on logistics performance.

Research Approach

It means that the present research is conducted within a framework of quantitative research design that utilizes empirical data for examining the relationship between transportation management and logistics performance. This helps to provide a reliable and valid measure of the association between variables in a more systematic and objective way. Mathematical interaction is possible to test hypotheses and establish causal relationships due to a quantitative design.

Research Design

The study employs an explanatory and descriptive design. The descriptive part identifies motivated transportation management practices of firms and the explanatory one looks at how those practices contribute to the logistics performance (outcomes). Such point-circles enable to observe and analyze relationships between parameters.

Sample and Data Collection

Research relies on primary and secondary data. A comprehensive questionnaire designed on Google forms was used to collect primary data. The questionnaire included close-ended question using a 5-point Likert scale (strongly disagree to strongly agree). The questions were designed to elicit respondents' opinions on critical components of transportation management – including sharing information, collaboration between partners, using technology and trust among them.

The Google Form generated from 50 responses in total. Our respondents were various types of employed professionals, unemployed, business owners and students, mostly are from Raipur, Chhattisgarh. This variety was useful in representing a breadth of perspectives on coordination practices and their relationships with business performance. Even though not all respondents work directly within logistics management, their familiarity with business operations adds an additional layer to the evaluation.

Data was collected from previous literatures in the form of textbooks, academic journals, online communities and papers released by their own company. These sources served as the basis for the theoretical framework of this study and to substantiate data analysis interpretation.

Variables of the Study

For research purpose, following variables have been taken as independent and dependent variable:

Independent Variables (Transportation Management Practices):

Route Optimization

Vehicle Utilization

Delivery Scheduling

Transportation Cost Control

Tracking Technology (e.g., GPS, TMS)

Dependent Variables (Logistics Performance Indicators):

Delivery Timeliness

Operational Efficiency

Service Quality

Cost Efficiency

Hypothesis Development

Based on the literature review and the objectives of the study, the following hypotheses were formulated to examine the relationship between transportation management practices and logistics performance:

H1: Transportation management practices have a significant positive impact on logistics performance.

H2: Adoption of tracking technology has a significant positive effect on logistics performance.

Data Analysis Technique

“In order to accomplish this objective, the use of descriptive statistics in this paper will be used as the tool for providing an organized interpretation of the data collected from this study.” Descriptive statistics is defined as a set of tools used in statistics to produce summaries of certain characteristics of the data collected that enable us to have a wholistic view of the respondents' opinion.

Since the research was conducted using a questionnaire, descriptive statistics was adopted as a tool for interpreting the raw data numerically. “Frequency and distribution statistics, along with percentiles, mean values and standard deviation, were used in describing statistics.” The demographics of the respondents were characterized using frequency and percent statistics for variables such as gender, age, and employment status.

Descriptive statistics was used to measure how respondents evaluate different aspects of transport management. “In this case, mean values were used for evaluating respondents' opinion on aspects such as route scheduling,

fleet management, timely deliveries, tracking systems, and expenses control.” Mean score presents general tendency towards the level of agreement/satisfaction/effectiveness related to particular variable.

The standard deviation measure is used to measure the degree of variability of the response from mean value. A low standard deviation indicates that the answers provided by respondents have the similarity (close to mean), while a high standard deviation suggests there is difference in opinions of those surveyed.

In addition to displaying all dependent variables, the use of visuals and descriptive statistics displayed some descriptors features that contributed towards certain levels of satisfaction. These were based on logistics performance indicators which involved service quality, operation, reliability and customer satisfaction. Thus, through the use of descriptive statistics, it was easier to establish general connection between transportation management and logistics performance.

Conceptual Research Model

The connection between the management of transportation and logistics performance can be seen through the conceptual model of the study. “In some recently published studies, it was found that such transportation management-related variables as route optimization, vehicle management, scheduling, cost management, and technology utilization may become independent regressors in connection with the logistic performance-related variables such as delivery efficiency, efficiency of operation, service quality, and cost efficiency.”

From this perspective, improvements in transportation management are expected to lead to better logistic performance and increased organizational efficiency and effectiveness.

Limitations of the Methodology

The research is conducted using some systematic approaches, but still some limitations could be seen. “First, convenience sampling could affect the validity of the results acquired. Second, respondents may have biases in their responses.” However, the study will help to understand the relationship between transport management and logistics performance in India.

DATA ANALYSIS AND FINDINGS

The analysis of the unprocessed data collected through a questionnaire conducted among 50 respondents is provided here exclusively. The analysis involves the use of descriptive statistics and graphs to study the connection between transportation management strategies and logistics performance.

Descriptive Statistics

In this study, descriptive analysis was employed to determine the perceptions of respondents concerning the transportation management processes and the level of logistics efficiency. “The results show that most respondents have positive perceptions regarding the efficiency of transportation systems to improve logistics efficiency.”

Tabel 5.1

Variable	Mean Score	Interpretation
Delivery Timeliness	3.85	High
Operational Efficiency	3.92	High
Service Quality	3.88	High
Cost Efficiency	3.45	Moderate
Route Optimization	3.90	High
Vehicle Utilization	3.75	Moderate to High
Delivery Scheduling	3.82	High
Tracking Technology	4.05	Very High

As reflected by the table, tracking technology with a Mean = 4.05 is ranked first which means that respondents definitely agree on using digital tools such as GPS and real time tracking systems. On the other hand, cost efficiency (Mean = 3.45) is the one with relatively lower score which reflect that still there are challenges exist related to transportation management on cost concerns.

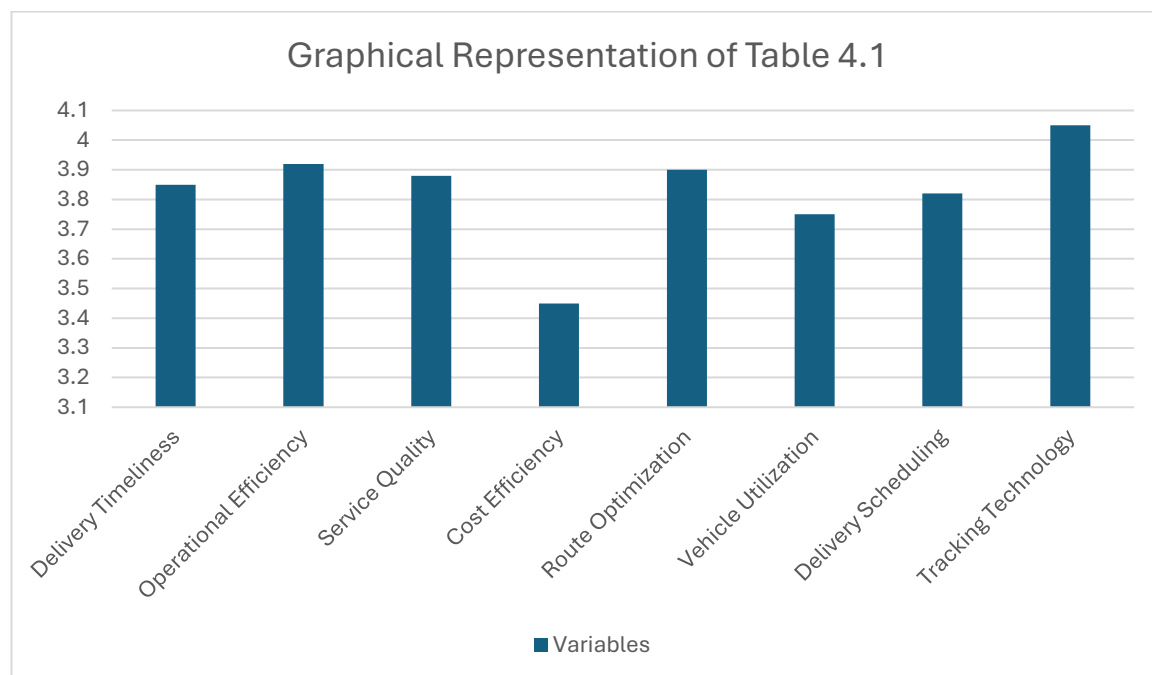
Graphical Representation

Hence, below you will find graphical representations depicting the mean scores of significant variables in this study to aid our understanding of the data and provide a more lucid visual interpretation of research results. Bar graphs and pie charts are also useful types of chart to provide a simple summary of large datasets. They assist in identifying trends, spotting differences between variables, as well as showing strengths and

weaknesses. The graphical tools were utilized to express the transportation management factors and the logistics performance indicators, which significantly increased clarity, readability and overall interpretation of research findings in this study.

Figure 4.1

Mean Scores of Key Variables — Bar Chart



The chart shows each of the variables visually for comparison. It provides a strong indication that the majority of transportation management practices received scores above (3) in average, reflecting satisfactory performance level. Adoption of new technologies is by far the most important factor.

Analysis of Transportation Management Practices

The mean score analysis shows that transportation management practices are an important factor in enhancing the logistics process. Tracking Technology was the highest variable in mean (4.05): logistically, the use of real-time tracking systems and digital visibility of shipment is becoming increasingly significant in the management of logistics. It therefore means that the integration of technology is considered highly significant in efforts to exercise control and transparency in transportation management activities.

Route optimization obtained a fairly high mean score (3.90), which indicated that the optimization of routes is an equally important activity that can help reduce delays in the process of delivery, distance covered, and fuel utilization. For example, delivery scheduling obtained a mean score of 3.82, implying that correct scheduling can guarantee timeliness and efficiency during the process of planning for deliveries. The second indicator, vehicle utilization (which is a measure of the efficiency of utilization of transportation assets) obtained a score of 3.75—a good but still moderate efficiency that requires improvements in areas such as asset management and planning.

In conclusion, key takeaways from the results of transportation management sanction the use of efficient planning systems tailored towards deliverables as well as proper utilization of available resources.

Impact on Logistics Performance

The study indicates a positive and highly significant relationship between transportation management practices and the logistics performance. Efficiencies gained in transportation lead to timely delivery services, enhancing the ability to serve customers well.

This can also mean efficient planning of routes and using resources so as not to create delays in logistics and make them effective. Again, the use of modern technological tools such as real-time tracking enhances the quality of the service provided.

On cost efficiency, it is slightly less compared to other performance criteria, meaning that organizations focus more on providing good-quality service and timeliness of their services than costs.

Key Findings

The third factor is that of transportation management, which, according to the research conducted, helps improve logistics performance in a supply chain. “Efficiency in transport management helps to ensure reliability of deliveries, efficiency of operations, and customer satisfaction. Of all the considered variables, technology use (specifically, GPS) proved to become the most significant determinant of assignment distances. Thus, there is no doubt about the wide use of information technology solutions for logistics.” Both routing and scheduling, which are examples of empirical variables, had their impact on deliveries, contributing to efficient performance by helping minimize possible delays and providing quality services. The importance of cost

management as a determinant of logistics performance has also been proven, although it is less significant compared to other variables. In general, efficient and technology-based transport management is a key determinant of logistics performance.

Interpretation of Results

The findings support previous studies that recognize transportation as an important variable in logistics chain management. Digital logistics revolution can be one such trend affecting the industry.

“Cost-related factors have been found to be less responsive, implying that some persistent barriers to Indian firms exist in terms of fuel prices, infrastructure, and networks.”

This finding supports earlier research since management of transportation is one of the key factors influencing the efficiency of logistics chain and ultimately organizational success.

Summary of Findings

The current study contributes to the existing body of knowledge since it substantiates the results that show the positive impact of the processes associated with transportation management on logistics performance. “The major determinants of effective logistics activities are the introduction of technology and route planning/scheduling. In other words, organizations should ensure that their performance is balanced through technology and efficiency gains.”

DISCUSSION

“The study results will ultimately add value to the existing body of knowledge in relation to the efficiency of transport management in enhancing organisational logistics performance in the context of India.” The study result confirms that transport management not only acts as a process management function, but also a strategic function for boosting efficiency and providing better services among others in the supply chain.

One of the key knowledge points gained from this research is that the application of higher technology levels substantially influences the pronouncement of logistics performance. “That should show the significance of changing logistics operations by relying on information technology. Knowing where things are right on the spot—and knowing your location in reference to the satellites as uS nimble "distance measurements"—say goodbye to uncertainty and factor in real-time monitoring.” It further enhances the decision making process at

each step of this process by providing information when, and when needed. This helps to assure delivery reliability, thereby satisfying customers. This is a new line of research that tackles the issue of visibility & flexibility in supply chains through tech.

Another important takeaway from this analysis is that routing optimization, and delivery speed, have a priceless role to play in organizational efficiency. On schedule routing and planning to ensure that scheduling visits to targets reduces travel distances and reduces fuel costs. Meanwhile, delivery time enables the timely delivery, which is very important concerning quality. “Like it makes an individual's tasks helpful when the timing is less daunting, and cuts the expectations of the recipient halfway when they're in for moves, so does this with transportation.”

“While this paper identifies another interesting finding, however, cost efficiency does not appear to be an important factor in the presence of factors of technology and others.” It suggests that companies are interested in offering quality service, not cut costs. Yes this appears to be a solid business move – because to restore and safeguard the moats – and ultimately to survive – is key to bring value.

Among logistic managers of course, there should be a managed transportation in any case from managerial point of view. The logistics performance needs to be maintained as far as technology, planning systems and resource allocation. “Efficiencies in organizations are expected to synchronize their transportation policy with objectives of their supply chain. The results prove even more significant for policymakers, as they highlighted the need to invest in the proper functioning of transportation infrastructure and the adoption of digital technologies in logistics operations.” Theoretically, New approaches such as National Logistics Policy and investment plans on transport infrastructure can help design improved roads of goods flows for resolving the Supply chain issues in India.

Generally speaking, the primary skills in transportation related planning and controlling for the various activities can be identified. “If scheduling methods in accordance with the market need comes into use and new logistics techniques software are produced, apart from timely provision of services, ecologically sustainable and competitive products are also produced.”

CONCLUSION AND FUTURE SCOPE

“The aim of the present study was to determine the role of such transportation management practices as route optimization, vehicle utilization, delivery scheduling, cost efficiency and technology adoption in delivering logistics performance.” The research has been validated through the empirical analysis of primary data extracted from visitors and the effect of transportation management on the logistics performance result.

Considering the results of the survey, transport is a significant determinant for the following areas: Delivery Timeliness, Operational Efficiency, and Service Quality. “The technology adoption turns out to be the outstanding variable toward one end of the spectrum in terms of real-time tracking system implementation among all the variables analyzed. At this stage of things, comfort in office jobs can be seen to be more prominent than other factors.” The optimization of the routes and the scheduling of the deliveries were also found to be the most important issues in the efficiency. This would prevent potential conflicts in regard to reduction in production or service output of entire truckloads at the installation stage, inhibition, or upgrades. The cost effect was relatively small compared to the effects of operations and technology factors, because it is part of transportation management efficiency. It signalled organizational transformations from cost cutting to enhancements of users' service qualities.

Theoretical Contribution

The study would be able to fill this void by providing the empirical evidence for the impact of transportation management on logistics performance for Indian sector. The paper highlights the importance of a supply chain and resource-based view by illustrating the potential competitive advantage of a single bundle of transportation capabilities.

Managerial Implications

The obtained results offer helpful information to logistics managers and practitioners. “Development of advance technology for transportation system, route planning as well as scheduling can contribute to effective logistics process within an organization. In fact, by applying one strategic planning strategy to the transportation system, you can achieve efficiency and customer satisfaction.”

However, the study has its limitations.

The research has some drawbacks though, despite its native contribution. “First of all, the data for the research was taken from different patients who were infected and admitted; however, the sample size is rather limited and hence the possibility of the results' generalization is possible only in the hospital environment.” Thirdly,

the present study uses self-reported data and thus may suffer from response bias. The use of only a portion of the factors and only selected aspects of transportation management is one limitation in the present analysis.

Future Research Scope

Future research work will be in this direction: analysis of the company level one of the ways. In addition, future studies may involve more advanced statistical techniques such as SEM which will allow describing variable relationships better. “Another aspect of future analysis may include exploring the influence of various innovative technologies such as AI and automation on volume of freight transportation.”

“Thus, it can be concluded that transportation management is a key component of logistics system, as well as a factor influencing organization's efficiency and productivity.” Taking electronic communication and effective management into account, in this highly competitive environment, organizations can realize sustainable development by elevating logistics performance.

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